Exploiting Latent Target Regimes: An EM Mixture-of-Regressors Approach to Ozone Concentration Prediction

Aamir Ahmad, Anisha Nanda, Aaryan Maheshwari, Pallav Rastogi, Sahil Pandey, Tattwa Shiwani
Indian Institute of Technology Kanpur

Abstract

Predicting atmospheric ozone (O_3) using low-cost electrochemical sensors is challenging due to inherent non-linearity and environmental cross-sensitivity. This report presents a comprehensive study of an Expectation-Maximization (EM) based mixture regression model. We demonstrate that by learning latent data partitions, the mixture model outperforms global linear approaches and standard Decision Trees. Specifically, a 4-component mixture, trained with 5 EM iterations, discovers contiguous target regimes that correlate with physical pollution levels. Our feature ablation highlights the critical role of temperature compensation in sensor accuracy. Finally, we provide a geometric interpretation of why mixture models provide superior interpolation compared to Decision Trees, quantifying the trade-off between predictive accuracy and inference latency.

1 Introduction

Low-cost air quality sensors are notorious for non-linear responses across varying pollution concentrations and ambient conditions. While a single Ridge regressor serves as a computationally efficient baseline, it assumes a global linear relationship that seldom exists in complex environmental data. Mixture models offer a middle ground between simple linear models and complex black-box non-linear regressors. By utilizing the Expectation-Maximization (EM) algorithm, we can learn a set of local linear regressors (components) and a routing mechanism. This study investigates the hyperparameters governing this architecture, the physical interpretation of learned components, and the structural advantages this approach holds over axis-aligned models like Decision Trees.

2 Methodology

The task involves predicting ground-truth `Ref. O3(ppb)` using five specific features: `Time` (hour-of-day), `Temp(C)`, `RH(%)`, `o3op1(mV)`, and `o3op2(mV)`. Following strict assignment constraints, no NO2-related features were utilized.

Data Split: The dataset (`DD1(Oct).csv`) was pre-processed to remove invalid entries, resulting in 10,458 samples. We employed a chronological split, using the first 80% (8,366 samples) for training and the final 20% (2,092 samples) for testing to simulate real-world temporal forecasting. All features were standardized based on training set statistics.

EM Algorithm for Mixed Regression:

1. **Initialization:** To avoid poor local minima, we partitioned the target variable into quantile-based bins and initialized component weights by fitting Ridge regressors ($\alpha = 0.01$) to each bin.

Preprint.

2. **E-step:** A "hard" assignment was performed where each training point was mapped to the component yielding the lowest absolute residual.

3. **M-step:** Each regressor was refit on its assigned data points using weighted Ridge regression.

4. **Inference (Auxiliary Classifier):** Since the true component assignment is unknown at test time, we trained a `LinearSVC` to predict component IDs from the input features. All reported Test Mean Absolute Errors (MAE) reflect the performance of this classifier-based routing (non-oracle).

3 Experiments & Results

3.1 Effect of EM Iterations

We fixed $n_{components} = 4$ and swept n_{iter} across 10 distinct values: $\{1, 2, 3, 5, 8, 12, 20, 35, 60, 100\}$. The objective was to identify the convergence point where the learned partitions stabilize.

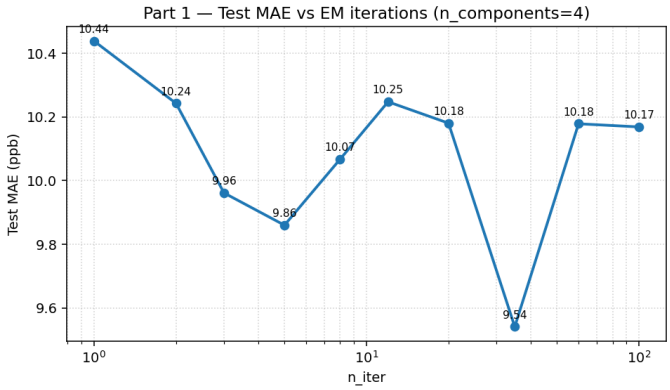


Figure 1: Test MAE (ppb) vs. EM Iterations ($n_{components} = 4$). The y-axis shows prediction error in ppb; the x-axis represents the number of training iterations. The curve represents the performance of the system using auxiliary classifier routing.

As shown in Figure 1, the Test MAE drops significantly during the first few steps, reaching a knee point at $n_{iter} = 5$ (MAE = 9.86 ppb). Beyond this, we observe non-monotonic "bouncing" behavior. This is an intuitive byproduct of hard EM: as the regressors refine, data points on the boundary of two regimes may flip-flop between components, causing the routing classifier to retrain on slightly different boundaries. This suggests that excessive iterations may lead to "boundary noise" rather than better generalization.

3.2 Effect of Number of Components

Using $n_{iter} = 5$, we evaluated mixture sizes: $n_{components} \in \{1, 2, 4, 8, 16, 32\}$.

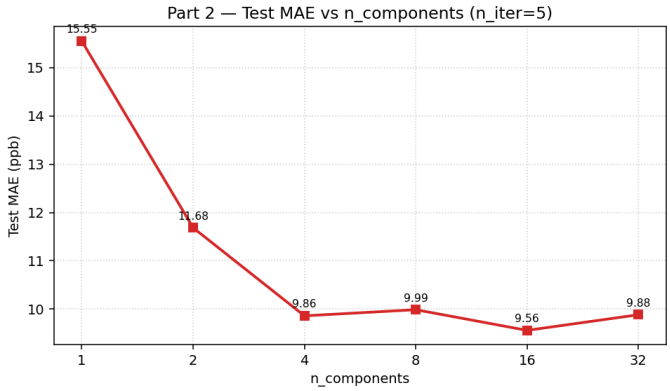


Figure 2: Test MAE (ppb) vs. Number of Components ($n_{components}$). The x-axis is on a base-2 log scale. The baseline at $n = 1$ represents a global linear Ridge regressor.

The transition from a global model ($n = 1$, MAE = 15.55 ppb) to a 4-component mixture (MAE = 9.86 ppb) yields a massive 37% error reduction. However, increasing components beyond 16 leads to diminishing returns and eventual degradation at $n = 32$ (MAE = 9.88 ppb). The intuition here is a classic bias-variance trade-off for the routing classifier: as the number of components increases, the classifier must learn increasingly complex and fragmented decision boundaries in a 5-dimensional feature space, eventually leading to routing errors that outweigh the benefits of localized linear fitting.

3.3 Interpretation of Components

By visualizing the distribution of ground-truth O_3 values assigned to each component, we gain insight into the model's logic.

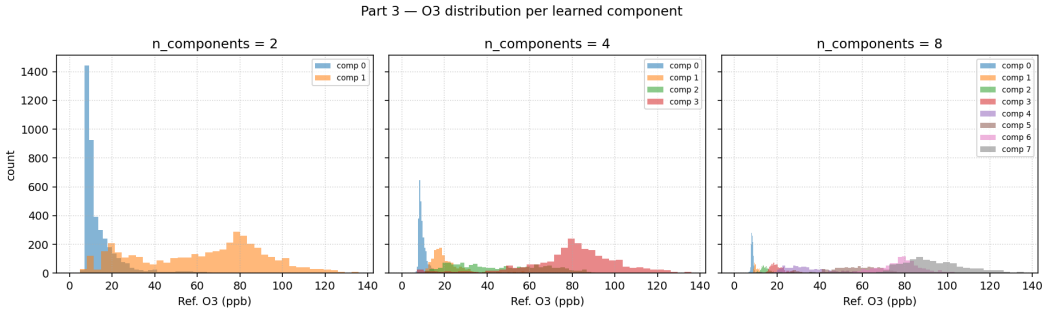


Figure 3: Target variable distribution (O3 in ppb) across learned components for $n = 2, 4$, and 8. The x-axis shows the ozone concentration; the y-axis shows the frequency of samples.

For $n = 4$, the EM algorithm discovers contiguous clusters corresponding to low (∼10 ppb), moderate (∼22 ppb), high-moderate (∼46 ppb), and extreme (∼80 ppb) pollution levels. This indicates that the non-linearity in the sensor is primarily a function of the target concentration itself—different "sensitivity modes" exist at different ozone levels. At $n = 8$, we observe significant overlap in the low-range distributions. This overlap is the causal factor for the MAE plateau: the routing classifier cannot distinguish between regimes with nearly identical feature signatures, leading to frequent misrouting.

3.4 Feature Ablation Study

We performed five leave-one-out ablations across the $n_{components}$ sweep to identify which environmental parameters are essential for the mixture model.

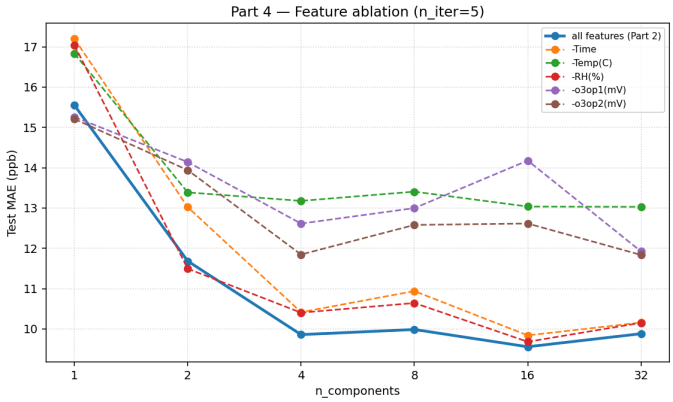


Figure 4: Ablation Study: Test MAE (ppb) vs. Number of Components. Each curve represents the model's performance when one specific feature is excluded.

The results in Figure 4 provide strong physical intuition:

- **Primary Drivers:** Removing `Temp(C)`, `o3op1`, or `o3op2` causes a catastrophic rise in MAE (to $\sim$13 ppb). Electochemcial sensors have a high temperature coefficient; without the `Temp` feature, the model cannot compensate for the baseline drift.
- **Redundancy:** Excluding `Time` or `RH(%)` results in minimal change (≤ 0.3 ppb). Since ozone levels and temperature are strongly correlated with the diurnal cycle, the `Temp` feature likely captures the information otherwise provided by `Time`.

3.5 Decision Tree vs Mixture Model

We benchmarked the $n = 4$ mixture model against a `DecisionTreeRegressor` tuned using internal validation on the training set (`min_samples_split=50`). We swept `max_depth` from 2 to 30 and measured inference time using `time.perf_counter()`.

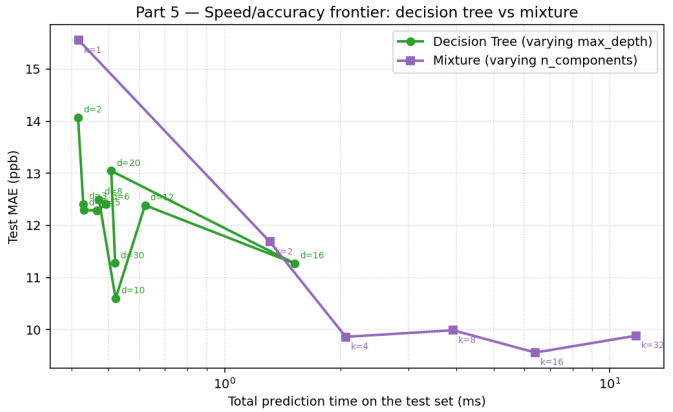


Figure 5: Speed-Accuracy Pareto Frontier: Test MAE (ppb) vs. Total Prediction Time (ms). Comparison between Decision Trees (varying depth) and the Mixture Model (varying components).

The mixture model ($n = 4$) achieves 9.86 ppb, outperforming the best Decision Tree (depth 10, 10.60 ppb) by 0.74 ppb. The intuition is geometric: A Decision Tree partitions the space into *axis-aligned hyper-rectangles* and predicts a *constant* value. The Mixture Model partitions the space into *affine half-spaces* (via SVC) and predicts an *affine function* (via Ridge). This allows the mixture model to interpolate trends within each region, whereas the DT is forced to approximate slopes with "stairs."

While the mixture model is $\sim 4\times$ slower due to the SVC and matrix-vector products, it provides a much smoother and more accurate response surface.

4 Discussion

The results suggest that mixed regression via EM is highly effective for sensor calibration. The algorithm's ability to discover latent pollution regimes (Part 3) allows it to bypass the need for manual non-linear feature engineering. A key insight is that while $n_{components}$ increases model capacity, the overall system accuracy is bottlenecked by the routing classifier's ability to identify those components at test time. For this specific sensor suite, a 4-component mixture provides a "sweet spot" where the regimes are physically distinct enough for a LinearSVC to route them correctly.

5 Conclusion

Our study concludes that a mixture-of-regressors approach is superior to both global linear models and Decision Trees for ozone prediction. With a configuration of $n_{components} = 4$ and $n_{iter} = 5$, the model achieves a test MAE of 9.86 ppb. We identified temperature and electrode voltages as the indispensable features for this task. While Decision Trees offer lower latency, the affine-leaf architecture of the mixture model provides a significant accuracy advantage, making it the preferred choice for high-precision environmental monitoring where a 2ms inference time is acceptable.